Dirichlet Multinomial Inference

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1 Dirichlet Multinomial Inference

Now we shall generalize the Beta-Binomial analysis to Dirichlet-Multinomial inference. First we shall look at the Multinomial distribution (which is a generalization of the Binomial distribution) and the Dirichlet distribution (which is a generalization of the Beta distribution).

1.1 Multinomial Distribution

The multinomial distribution is a generalization of the Binomial distribution. We first recall the binomial distribution. We say that $X \sim \text{Bin}(n, p)$ if

$$P\{X = x\} = \frac{n!}{x!(n-x)!} p^x (1-p)^{n-x} \quad \text{for } x = 0, 1, \dots, n.$$

X represents the count of successess in n repetitions of a simple binary experiment with two outcomes success (probability p) and failure (probability $1-p$).

The multinomial distribution is obtained by considering n repetitions of an experiment with k outcomes (for some $k \geq 1$) with probabilities $p_1, \dots, p_k$ (we need $p_i \geq 0$ and $\sum_{i=1}^k p_i = 1$). Let X_i denote the count of the i-th outcome among the n repetitions of the experiment (in other words, X_i is the number of times the i-th outcome happens in the n trials). The joint distribution of $(X_1, \dots, X_k)$ is written as Multinomial$(n; p_1, \dots, p_k)$. It corresponds to the probabilities:

$$P\{X_1 = x_1, \dots, X_k = x_k\} = \begin{cases} \dfrac{n!}{x_1! \cdots x_k!} \displaystyle\prod_{i=1}^k p_i^{x_i}, & \text{if } x_i \geq 0 \text{ for all } i \text{ and } \sum_{i=1}^k x_i = n, \\ 0, & \text{otherwise.} \end{cases}$$

The following consequences of $(X_1, \dots, X_k) \sim \text{Multinomial}(n; p_1, \dots, p_k)$ are straightforward to see:

1. To derive the distribution of X_i i.e., $P\{X_i = x\}$, we can write

$$\begin{aligned} P\{X_i = x\} &= P\{x \text{ trials had outcome } i, (n-x) \text{ trials had outcome not } i\} \\ &= \frac{n}{x!(n-x)!} p_i^x (1-p_i)^{n-x}. \end{aligned}$$

   In other words $X_i \sim \text{Bin}(n, p_i)$. In essence, here we consider the variant of the original experiment where, for each trial, we just record if the original outcome was 'not i' or i. This converts the setup to binary trials and X_i represents the number of successes (success = outcome is i) so $X_i \sim \text{Bin}(n, p_i)$.

2. The joint distribution of (X_i, X_j) (for fixed $i \neq j$) can be derived in the following way:

$$\begin{aligned} &P\{X_i = x_1, X_j = x_2\} \\ &= P\{x_1 \text{ trials were } i, x_2 \text{ trials were } j, n - x_1 - x_2 \text{ trials were neither } i \text{ nor } j\} \\ &= \frac{n!}{x_1! x_2! (n - x_1 - x_2)!} p_i^{x_1} p_j^{x_2} (1 - p_i - p_j)^{n - x_1 - x_2} \end{aligned}$$

   provided $x_1, x_2 \in \{0, 1, \dots, n\}$ with $x_1 + x_2 \leq n$ (otherwise, the probability equals 0).

3. It can be checked (see e.g., https://en.wikipedia.org/wiki/Multinomial_distribution) that

$$\mathbb{E}X_i = np_i, \quad \text{var}(X_i) = np_i(1-p_i), \text{ and } \quad \text{cov}(X_i, X_j) = -np_ip_j \text{ if } i \neq j.$$

1.2 Dirichlet Distribution

The Dirichlet distribution is a generalization of the Beta distribution, see Gelman et al. [2013, Chapter 3]). Recall the Beta(a, b) distribution is a distribution over $p \in [0, 1]$ corresponding to the density:

$$f(p) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} p^{a-1}(1-p)^{b-1} I\{0 \leq p \leq 1\}.$$

This density function is well-defined only when both a and b are strictly positive. However one can still define the Beta(a, b) distribution even when one or both of a and b are zero. This can be done by a limiting argument:

1. Beta$(0, 0) = \lim_{\epsilon \downarrow 0}$ Beta(ϵ, ϵ) = Bernoulli(0.5). In other words, Beta$(0, 0)$ becomes a discrete two-point distribution assigning equal probability to 0 and 1.

2. For fixed $b > 0$, we have Beta$(0, b) = \lim_{\epsilon \downarrow 0}$ Beta(ϵ, b) = Bernoulli$(0) = \delta_{\{0\}}$. In other words, Beta$(0, b)$ is the point mass as 0.

3. For fixed $a > 0$, we have Beta$(a, 0) = \lim_{\epsilon \downarrow 0}$ Beta(a, ϵ) = Bernoulli$(1) = \delta_{\{1\}}$. In other words, Beta$(a, 0)$ is the point mass as 1.

The limiting statements above can be made rigorous by moment calculations (e.g., by showing that the moments of Beta(ϵ, ϵ) converge to the moments of Bernoulli(0.5) as $\epsilon \downarrow 0$).

The Beta distribution can be viewed as the distribution over the probabilities (p and $1-p$) of an experiment with two outcomes. The Dirichlet distribution is the distribution over the probabilities $p_1, \dots, p_k$ of an experiment with k outcomes. Specifically given $a_1, \dots, a_k$, the Dirichlet$(a_1, \dots, a_k)$ distribution corresponds to the density:

$$\frac{\Gamma(a_1 + \cdots + a_k)}{\Gamma(a_1) \dots \Gamma(a_k)} p_1^{a_1 - 1} \dots p_k^{a_k - 1} I\{p_1, \dots, p_k \geq 0, \sum_i p_i = 1\}.$$

Note that this should be viewed as density on $(k-1)$-dimensional space (as opposed to k-dimensional space) because of the restriction $p_1 + \dots p_k = 1$. In other words, for every $A \subseteq^k$, we have

$$\begin{aligned}
&P\{(p_1, \dots, p_k) \in A\} \\
&= \frac{\Gamma(a_1 + \cdots + a_k)}{\Gamma(a_1) \dots \Gamma(a_k)} \int_S p_1^{a_1 - 1} \dots p_{k-1}^{a_{k-1} - 1} (1 - p_1 - \cdots - p_{k-1})^{a_k - 1} \\
&I\{(p_1, \dots, p_{k-1}, 1 - p_1 - \cdots - p_{k-1}) \in A\} dp_1 \dots dp_{k-1}
\end{aligned}$$

where

$$S := \{(p_1, \dots, p_{k-1}) : p_i \geq 0, p_1 + \cdots + p_{k-1} \leq 1\}.$$

It can be checked that if $(p_1, \dots, p_k) \sim$ Dirichlet$(a_1, \dots, a_k)$, then (see e.g., https://en.wikipedia.org/wiki/Dirichlet_distribution)

$$\begin{aligned}
\mathbb{E} p_i &= \frac{a_i}{a_1 + \cdots + a_k} \\
\mathrm{var}(p_i) &= \left(\frac{a_i}{a_1 + \cdots + a_k}\right) \left(\frac{(a_1 + \cdots + a_k) - a_i}{a_1 + \cdots + a_k}\right) \left(\frac{1}{a_1 + \cdots + a_k + 1}\right) \\
\mathrm{cov}(p_i, p_j) &= -\left(\frac{a_i}{a_1 + \cdots + a_k}\right) \left(\frac{a_j}{a_1 + \cdots + a_k}\right) \left(\frac{1}{a_1 + \cdots + a_k + 1}\right).
\end{aligned}$$

Remark 1 (Connection with the Gamma distribution)**.** *A particularly useful way to construct the Dirichlet distribution is through independent Gamma random variables. Let*

$$G_i \overset{ind}{\sim} \text{Gamma}(a_i, 1), \qquad i = 1, \dots, k,$$

where the Gamma density is

$$f(x) = \frac{1}{\Gamma(a_i)} x^{a_i - 1} e^{-x}, \qquad x > 0.$$

Define

$$T = G_1 + \cdots + G_k, \qquad P_i = \frac{G_i}{T}, \qquad i = 1, \dots, k.$$

Then

$$(P_1, \dots, P_k) \sim \text{Dirichlet}(a_1, \dots, a_k),$$

and the random variable T is independent of $(P_1, \dots, P_k)$ with

$$T \sim \text{Gamma}(a_1 + \cdots + a_k, 1).$$

Thus the Dirichlet distribution can be interpreted as the distribution obtained by taking independent Gamma random variables and normalizing them so that they sum to one. This representation explains why the Dirichlet distribution is supported on the probability simplex, and is also the basis of the simplest algorithm for generating Dirichlet random vectors in practice. The Beta distribution is the special case $k = 2$. Indeed, if

$$G_1 \sim \text{Gamma}(a, 1), \qquad G_2 \sim \text{Gamma}(b, 1),$$

are independent, then

$$\frac{G_1}{G_1 + G_2} \sim \text{Beta}(a, b).$$

As for the case of the Beta, the following facts about the Dirichlet can also be proved using moment calculations:

1. $\text{Dirichlet}(0, \dots, 0) = \lim_{\epsilon \downarrow 0} \text{Dirichlet}(\epsilon, \dots, \epsilon)$ is the discrete uniform distribution on $e_1 \dots, e_k$ with e_i is the vector which has 1 in the i-th location and 0 everywhere else i.e.,

$$\text{Dirichlet}(0, \dots, 0) = \frac{1}{k} \sum_{i=1}^{k} \delta_{\{e_i\}}.$$

2. If $k = 4$ and $a_2, a_4 > 0$, then $\text{Dirichlet}(0, a_2, 0, a_4) = \lim_{\epsilon \downarrow 0} \text{Dirichlet}(\epsilon, a_2, \epsilon, a_4)$ is supported on $\{(p_1, p_2, p_3, p_4) : p_1 = 0, p_2 \geq 0, p_3 = 0, p_4 \geq 0, p_1 + p_2 + p_3 + p_4 = 1\}$, and the distribution of (p_2, p_4) is $\text{Dirichlet}(a_2, a_4)$. Informally, we write

$$\text{Dirichlet}(0, a_2, 0, a_4) = \text{Dirichlet}(a_2, a_4).$$

3. More generally, let $a = (a_1, \dots, a_k)$ with each $a_i \geq 0$. Let $I = \{i : a_i > 0\}$ and let m be the cardinality of I. Then $\text{Dirichlet}(a) := \lim_{\epsilon \downarrow 0} \text{Dirichlet}(a_1 + \epsilon, \dots, a_k + \epsilon)$ satisfies the following.

   (a) The support of $\text{Dirichlet}(a)$ equals $\{(p_1, \dots, p_k) : p_i \geq 0, \sum_{i=1}^{k} p_i = 1, p_i = 0 \text{ for all } i \notin I\}$.

   (b) If $p \sim \text{Dirichlet}(a)$, then the subvector $p_i, i \in I$ satisfies:

$$(p_i, i \in I) \sim \text{Dirichlet}(a_i, i \in I).$$

   Note that the above is a Dirichlet distribution for an m-dimensional probability vector.

1.3 Dirichlet Prior and Multinomial Likelihood

We saw the following basic fact:

$$p \sim \text{Beta}(a, b) \quad \text{and} \quad X \mid p \sim \text{Bin}(n, p) \implies p \mid X = x \sim \text{Beta}(a + x, b + n - x).$$

The generalization of this is:

$$(p_1, \dots, p_k) \sim \text{Dirichlet}(a_1, \dots, a_k) \text{ and } (X_1, \dots, X_k) \mid (p_1, \dots, p_k) \sim \text{Multinomial}(n; p_1, \dots, p_k)$$
$$\implies (p_1, \dots, p_k) \mid X_1 = x_1, \dots, X_k = x_k \sim \text{Dirichlet}(a_1 + x_1, a_2 + x_2, \dots, a_k + x_k).$$

So the posterior mean estimate of p_i is given by:

$$\mathbb{E}(p_i \mid X_1 = x_1, \dots, X_k = x_k) = \frac{a_i + x_i}{(a_1 + x_1) + \cdots + (a_k + x_k)} = \frac{a_i + x_i}{n + a_1 + \cdots + a_k}$$

where we used $x_1 + \cdots + x_k = n$.

Remark 2. *The above posterior characterization is true for any $a_1, \dots, a_k$ and then one can also make sense of the statement for all $a_1, \dots, a_k$ equal to 0 by taking limits.*

1.4 Special Case: the Dirichlet$(0, \dots, 0)$ prior

In the special case where all $a_i = 0$, then the posterior mean of p_i is simply x_i/n; so we are estimating the probability of the i-th outcome by simply the proportion of times the i-th outcome appeared in the n trials. This is just the frequentist MLE. So the posterior mean estimate coincides with the frequentist MLE when the prior is Dirichlet$(0, \dots, 0)$.

More precisely, the posterior for the Dirichlet$(0, \dots, 0)$ prior is Dirichlet$(x_1, \dots, x_k)$. A key property of this posterior is that if $x_i = 0$ for some i, then the posterior places all its mass on the set $\{p_i = 0\}$. In other words, any outcome that is not observed in the sample is assigned zero probability under the posterior and is therefore completely excluded from future consideration.

As an explicit example, suppose $k = 6$, $n = 4$, $x_1 = 0, x_2 = 0, x_3 = 0, x_4 = 0, x_5 = 3, x_6 = 1$. In other words, we rolled a six-faced die 4 times; in 3 of the 4 rolls, we observed the outcome 5, and in the other roll, we observed a 6. After observing this data (note that we started with the prior Dirichlet$(0, 0, 0, 0, 0, 0)$), our posterior becomes Dirichlet$(0, 0, 0, 0, 3, 1)$. This means that the posterior is concentrated on the probabilities $(0, 0, 0, 0, p_5, 1 - p_5)$ with $p_5 \sim \text{Beta}(3, 1)$.

Remark 3 (Interpretation of the Dirichlet$(0, \dots, 0)$ prior)**.** *The prior* Dirichlet$(0, \dots, 0)$ *has the attractive property that the posterior mean coincides exactly with the frequentist maximum likelihood estimator:*

$$E(p_i \mid X) = \frac{x_i}{n}.$$

Thus the prior contributes no pseudo-counts to the posterior and, in this limited sense, behaves as an "uninformative" prior.

It is important to note, however, that this prior is not noninformative in the usual Bayesian sense. Indeed,

$$\text{Dirichlet}(0, \dots, 0) = \lim_{\varepsilon \downarrow 0} \text{Dirichlet}(\varepsilon, \dots, \varepsilon)$$

is the discrete uniform distribution on the vertices of the probability simplex. Thus, before observing any data, the prior assigns equal probability to the hypotheses

$$(1, 0, \dots, 0),\ (0, 1, \dots, 0),\ \dots,\ (0, \dots, 0, 1),$$

corresponding to the belief that exactly one outcome has probability one, although there is complete uncertainty as to which outcome this is. Consequently, if an outcome is never observed, its posterior probability is forced to be zero. This explains why the posterior mean coincides with the maximum likelihood estimator while simultaneously illustrating that the prior is highly informative from a geometric point of view.

Remark 4. *Something seems fishy here. The prior Dirichlet*$(0,0,0,0,0,0)$*) is supported on the vertices of the probability simplex. However, the posterior seems to be nonzero even in the interior. We have to be careful in how we interpret this. The correct way to interpret this is that we start with the prior Dirichlet*$(\epsilon,\dots,\epsilon)$*) and obtain the posterior and then take the limit as* $\epsilon \to 0$.

2 The Bayesian Bootstrap

Suppose that the sample space consists of a finite alphabet

$$\Omega = \{1, \dots, K\}, \tag{1}$$

and let

$$p = (p_1, \dots, p_K) \tag{2}$$

denote the unknown probability vector. We observe independent samples

$$Y_1, \dots, Y_n \overset{\text{i.i.d.}}{\sim} p. \tag{3}$$

Let

$$X_j = \sum_{i=1}^{n} \mathbf{1}\{Y_i = j\} \tag{4}$$

denote the number of times the symbol j appears in the sample. Then

$$(X_1, \dots, X_K) \sim \text{Multinomial}(n; p_1, \dots, p_K). \tag{5}$$

A natural Bayesian prior on the probability vector p is the Dirichlet distribution. If

$$p \sim \text{Dirichlet}(a_1, \dots, a_K), \tag{6}$$

then conjugacy implies that

$$p \mid X \sim \text{Dirichlet}(a_1 + X_1, \dots, a_K + X_K). \tag{7}$$

The posterior is a probability distribution on the simplex of all probability vectors. Equivalently, we may regard it as a probability distribution on probability measures through the representation

$$P = \sum_{j=1}^{K} p_j \delta_j, \tag{8}$$

where δ_j denotes the point mass at the symbol j.

Now suppose that all observations are distinct:

$$Y_1, \dots, Y_n \text{ are all distinct.} \tag{9}$$

This can only happen when $K \geq n$. In this case every observed symbol appears exactly once, so the count vector takes the form (by renaming the symbols)

$$(1, \ldots, 1, 0, \ldots, 0). \tag{10}$$

Consider the improper prior

$$\text{Dirichlet}(0, \ldots, 0). \tag{11}$$

The posterior is then

$$\text{Dirichlet}(1, \ldots, 1, 0, \ldots, 0). \tag{12}$$

Consequently, every unobserved symbol receives posterior probability zero. Infact, we can represent the (random) posterior probability measure as

$$P^{posterior} = \sum_{j=1}^{n} W_j \delta_{Y_j}.$$

where

$$(W_1, \ldots, W_n) \sim \text{Dirichlet}(1, \ldots, 1). \tag{13}$$

Since the density of the Dirichlet$(1, \ldots, 1)$ distribution is constant, this is simply the uniform distribution on the $(n-1)$-dimensional simplex.

The support points are fixed at the observed data values, while the uncertainty is represented entirely through the random weights.

Posterior samples therefore can be generated by generating:

$$(w_1^{(j)}, \ldots, w_n^{(j)}) \overset{\text{i.i.d}}{\sim} \text{Dirichlet}(1, \ldots, 1)$$

for $j = 1, \ldots, M$ (for a large number M). Inference done with these samples is referred to as the **Bayesian Bootstrap** (see Rubin [1981]). This is because the generation process for these samples is similar to the usual nonparametric bootstrap sample generation. Indeed, the usual bootstrap samples can be seen as being generated from:

$$\sum_{i=1}^{n} v_i \delta_{\{y_i\}} \quad \text{where } (v_1, \ldots, v_n) \sim \frac{1}{n}\text{Multinomial}(n; 1/n, \ldots, 1/n).$$

The distributions $(w_1, \ldots, w_n) \sim \text{Dir}(1, \ldots, 1)$ & $(v_1, \ldots, v_n) \sim \text{Mult}(n; 1/n, \ldots, 1/n)/n$ are quite close when n is large. For example,

$$\begin{aligned}
&\mathbb{E}w_i = \mathbb{E}v_i = 1/n \\
&\text{var}(w_i) = \frac{n-1}{n^2(n+1)} \approx \text{var}(v_i) = \frac{n-1}{n^3} \\
&\text{correlation}(w_i, w_j) = \text{correlation}(v_i, v_j) = \frac{-1}{n-1} \quad \text{when } i \neq j.
\end{aligned}$$

So operationally, bootstrap and the Bayesian bootstrap give very similar results, although conceptually they are quite different.

The two procedures are also conceptually different in how they are used for statistical inference. In the classical bootstrap, one first generates bootstrap samples, computes the statistic of interest on each bootstrap sample, and then uses the empirical distribution of these bootstrap replicates to approximate the sampling distribution of the estimator. In particular, if the goal is to estimate the variance of an estimator, one simply computes the sample variance of the bootstrap replicates.

In the Bayesian bootstrap, the Dirichlet samples correspond to draws from the posterior distribution of the unknown probability measure P. Suppose the parameter of interest is a functional $\theta(P)$, such as the mean, median, variance, or a regression coefficient. For each posterior sample

$$P^{(j)} = \sum_{i=1}^{n} w_i^{(j)} \delta_{\{y_i\}},$$

we compute

$$\theta^{(j)} = \theta(P^{(j)}), \qquad j = 1, \ldots, M.$$

The empirical distribution of

$$\theta^{(1)}, \ldots, \theta^{(M)}$$

then approximates the posterior distribution of $\theta(P)$. Posterior means, posterior standard deviations, credible intervals, and other Bayesian summaries can be obtained directly from these simulated values.

Thus, although the computational procedures are similar, the inferential targets are fundamentally different. The classical bootstrap uses Monte Carlo simulation to approximate the frequentist sampling distribution of an estimator, whereas the Bayesian bootstrap uses Monte Carlo simulation to approximate the Bayesian posterior distribution of the parameter, or functional, of interest.

For further discussion of the relationship between the two procedures, see Rubin [1981]. For more information on the Bayesian bootstrap and its relation to the classical bootstrap, see the blog post by Rasmus Bååth: https://www.sumsar.net/blog/2015/04/the-non-parametric-bootstrap-as-a-baye

3 Bootstrap Confidence and Credible Intervals

Suppose θ is an unknown parameter and $\hat{\theta}$ is an estimator computed from the observed data.

3.1 The Basic Bootstrap Confidence Interval

The goal is to construct a confidence interval for the unknown parameter θ. Ideally, we would like to know the sampling distribution of the estimation error

$$\hat{\theta} - \theta.$$

Since this distribution depends on the unknown population, it is typically unavailable.

The classical bootstrap approximates this distribution by repeatedly sampling with replacement from the observed data. Specifically, generate B bootstrap samples. For the b-th bootstrap sample, compute the corresponding estimate

$$\hat{\theta}^{*(b)}, \qquad b = 1, \ldots, B.$$

The fundamental bootstrap approximation is

$$\hat{\theta} - \theta \approx \hat{\theta}^{*} - \hat{\theta},$$

where $\hat{\theta}^{*}$ denotes the estimator computed from a bootstrap sample.

Let

$$q_{0.025} \quad \text{and} \quad q_{0.975}$$

denote the empirical 2.5% and 97.5% quantiles of the bootstrap replicates

$$\hat{\theta}^{*(1)}, \hat{\theta}^{*(2)}, \ldots, \hat{\theta}^{*(B)}.$$

Rearranging the approximation

$$\hat{\theta} - \theta \approx \hat{\theta}^* - \hat{\theta}$$

gives

$$\theta \approx 2\hat{\theta} - \hat{\theta}^*.$$

Consequently, an approximate 95% confidence interval for θ is

$$\boxed{\left[2\hat{\theta} - q_{0.975},\, 2\hat{\theta} - q_{0.025}\right].}$$

This interval is called the *basic bootstrap confidence interval.* Its interpretation is the same as that of any frequentist confidence interval: under repeated sampling from the population, the procedure produces an interval containing the true parameter approximately 95% of the time.

3.2 The Bayesian Bootstrap Credible Interval

Recall that the Bayesian bootstrap places a Dirichlet$(1, \ldots, 1)$ prior on the unknown probability weights assigned to the observed data. Thus,

$$(w_1, \ldots, w_n) \sim \text{Dirichlet}(1, \ldots, 1),$$

where

$$w_i \geq 0, \qquad \sum_{i=1}^{n} w_i = 1.$$

Suppose $\hat{\theta}(w_1, \ldots, w_n)$ denotes the estimator computed using these weights. The Bayesian bootstrap proceeds as follows.

1. Generate independent weight vectors

$$(w_1^{(b)}, \ldots, w_n^{(b)}) \sim \text{Dirichlet}(1, \ldots, 1), \qquad b = 1, \ldots, B.$$

2. For each set of weights, compute the corresponding weighted estimate

$$\hat{\theta}^{(b)} = \hat{\theta}(w_1^{(b)}, \ldots, w_n^{(b)}).$$

3. The collection

$$\hat{\theta}^{(1)}, \hat{\theta}^{(2)}, \ldots, \hat{\theta}^{(B)}$$

   forms an approximate sample from the posterior distribution of θ.

A 95% Bayesian credible interval is obtained by taking the empirical 2.5% and 97.5% quantiles of the posterior samples.

Unlike the classical bootstrap, no approximation to a sampling distribution or inversion argument is required. The Bayesian bootstrap produces approximate posterior samples directly, so the credible interval is simply obtained by taking posterior quantiles. Consequently, the Bayesian bootstrap admits a direct probabilistic interpretation: conditional on the observed data, the posterior probability that θ lies in the credible interval is approximately 95%.

4 Simple Language Models

We will illustrate Dirichlet-Multinomial inference on a text data example. This analysis is taken from MacKay and Peto [1995]. We observe some text $y_1, \dots, y_n$. The vocabulary consists of the number of distinct words (including punctuation marks) in the text. Assume the vocabulary size is k.

4.1 A simple i.i.d model

The simplest model assumes that $y_1, \dots, y_n$ are i.i.d with an distribution P that is supported on the vocabulary. The likelihood then becomes:

$$\prod_{i=1}^{n} p_{y_i} = \prod_{i=1}^{k} p_i^{N_i} \tag{14}$$

where N_i is the observed count for the i-th word in the vocabulary. We can place a Dirichlet prior on $(p_1, \dots, p_k)$ (e.g., Dirichlet$(0, \dots, 0)$) which would result in the Dirichlet$(N_1, \dots, N_k)$ posterior distribution. One can then generate new text in the following way:

1. First generate $(p_1, \dots, p_k)$ from the posterior distribution.
2. Then generate new words $y_{n+1}, y_{n+2}, \dots$ i.i.d from the multinomial distribution Mult$(1; p_1, \dots, p_k)$ until a specific word (e.g., the period symbol '.')) is generated.

Obviously, this will not result in realistic text. This is because the i.i.d assumption collapses the text into a bag-of-words representation, discarding all sequential information. Indeed, in this model, we are effectively working with the counts $N_1, \dots, N_k$ instead of the raw data $y_1, \dots, y_n$. The likelihood in terms of the counts is:

$$\frac{n}{N_1! \dots N_k!} p_1^{N_1} \dots p_k^{N_k} \propto \prod_{i=1}^{k} p_i^{N_i}$$

which coincides with (14). As a result, it cannot capture even the most basic linguistic dependencies (e.g., that "the" is likely to be followed by a noun).

4.2 AR(1) model

A slightly better model will be obtained if we replace the i.i.d assumption with a Markovian one. This is also referred to as the first order AutoRegressive AR(1) model.

The likelihood here then becomes:

$$p_{y_1} p_{y_2|y_1} p_{y_3|y_2,y_1} \cdots p_{y_n|y_{n-1},\dots,y_1}$$

We assume now that $p_{y_i|y_{i-1},\dots,y_1} = p_{y_i|y_{i-1}}$ for each $i = 3, \dots, n$. This gives

$$\text{Likelihood} = p_{y_1} \prod_{i=2}^{n} p_{y_i|y_{i-1}}.$$

We also assume that p_{y_1} is constant so we don't need to model it. This gives:

$$\text{Likelihood} \propto \prod_{i=2}^{n} p_{y_i|y_{i-1}}. \tag{15}$$

If you don't like the assumption that p_{y_1} is constant, you can interpret the above as the conditional likelihood of $(y_2, \dots, y_n)$ given y_1.

Just as in the i.i.d model where we could write the likelihood in terms of counts, we can rewrite the likelihood (15) using 'bi-gram counts'. For each i and j in $\{1, \dots, k\}$, let $x_{j|i}$ denote the number of times the word j follows word i in the text. Also let $x_i = \sum_{j=1}^{k} x_{j|i}$ denote the number of times word i appears as the "previous word" in the text.

Then the likelihood (15) is the same as:

$$\prod_{i=1}^{k} \prod_{j=1}^{k} p_{j|i}^{x_{j|i}} \tag{16}$$

where $p_{j|i}$ denotes the probability that word j appears as the next word given that the previous word is i.

We can also directly write the likelihood in terms of the bigram counts $(x_{j|i}, j = 1, \dots, k)$ and $i = 1, \dots, k$. The model here is:

$$(x_{j|i}, j = 1, \dots, k) \overset{\text{ind}}{\sim} \text{Multinomial}(x_i; p_{j|i}, j = 1, \dots, k).$$

so that the likelihood in terms of the bigram counts is:

$$\prod_{i=1}^{k} \frac{x_i!}{\prod_{j=1}^{k} x_{j|i}!} \prod_{j=1}^{k} p_{j|i}^{x_{j|i}}. \tag{17}$$

The unknown parameters in this model are the probability vectors $(p_{j|i}, 1 \leq j \leq k)$ for each $i = 1, \dots, k$.

MacKay and Peto [1995] perform Bayesian inference with the following prior:

$$(p_{j|i}, j = 1, \dots, k) \overset{\text{i.i.d}}{\sim} \text{Dirichlet}(a_1, \dots, a_k)$$

for some $a_1, \dots, a_k > 0$. This means that all the k probability vectors $(p_{j|i}, 1 \leq j \leq k)$ are assumed to be independently distributed according to the same Dirichlet distribution Dirichlet$(a_1, \dots, a_k)$. The prior density is therefore

$$\prod_{i=1}^{k} \frac{\Gamma(a_1 + \dots + a_k)}{\Gamma(a_1) \dots \Gamma(a_k)} \prod_{j=1}^{k} p_{j|i}^{a_j - 1}.$$

The posterior is then:

$$\prod_{i=1}^{k} \frac{x_i!}{\prod_{j=1}^{k} x_{j|i}!} \prod_{j=1}^{k} p_{j|i}^{x_{j|i}} \times \prod_{i=1}^{k} \frac{\Gamma(a_1 + \dots + a_k)}{\Gamma(a_1) \dots \Gamma(a_k)} \prod_{j=1}^{k} p_{j|i}^{a_j - 1} \propto \prod_{i=1}^{k} \prod_{j=1}^{k} p_{j|i}^{x_{j|i} + a_j - 1}$$

which means that:

$$(p_{j|i}, j = 1, \dots, k) \overset{\text{ind}}{\sim} \text{Dirichlet}(x_{j|i} + a_j, j = 1, \dots, k).$$

So the posterior mean estimate of $p_{j|i}$ is:

$$\hat{p}_{j|i} = \frac{x_{j|i} + a_j}{x_i + a_1 + \dots + a_k}.$$

The choice of $a_1, \dots, a_k$ controls the properties of the above posterior. Instead of plugging in some values for $a_1, \dots, a_k$, MacKay and Peto [1995] propose estimating them by maximizing the marginal likelihood of the data $(x_{j|i}, 1 \le i, j \le k)$ over all values of $a_1, \dots, a_k$.

The marginal distribution of the data given the Dirichlet hyperparameters $a_1, \dots, a_k$ is:

$$
\begin{aligned}
&P\left\{x_{j|i}, 1 \le i, j \le k \mid a_1, \dots, a_k\right\} \\
&= \int \prod_{i=1}^{k} \frac{x_i!}{\prod_{j=1}^{k} x_{j|i}!} \prod_{j=1}^{k} p_{j|i}^{x_{j|i}} \times \prod_{i=1}^{k} \frac{\Gamma(a_1 + \dots + a_k)}{\Gamma(a_1) \dots \Gamma(a_k)} \prod_{j=1}^{k} p_{j|i}^{a_j - 1} d(p_{j|i} \text{ all } j, i) \\
&= \prod_{i=1}^{k} \frac{x_i!}{\prod_{j=1}^{k} x_{j|i}!} \frac{\Gamma(a_1 + \dots + a_k)}{\Gamma(a_1) \dots \Gamma(a_k)} \int \prod_{j=1}^{k} p_{j|i}^{x_{j|i} + a_j - 1} d(p_{j|i}, 1 \le j \le k) \\
&= \prod_{i=1}^{k} \frac{x_i!}{\prod_{j=1}^{k} x_{j|i}!} \left[\frac{\Gamma(a_1 + \dots + a_k)}{\Gamma(x_i + a_1 + \dots + a_k)} \prod_{j=1}^{k} \frac{\Gamma(x_{j|i} + a_j)}{\Gamma(a_j)}\right].
\end{aligned}
$$

Remark 5. *The normalizing constant of the Dirichlet is explicit in terms of Gamma integrals. This can again be proved by considering connections to the gamma distribution.*

In the machine learning literature, the marginal distribution of the data given the hyperparameters of the prior is referred to as the Evidence (see e.g., https://en.wikipedia.org/wiki/Marginal_likelihood). So:

$$
\text{Evidence}(a_1, \dots, a_k) = \prod_{i=1}^{k} \frac{x_i!}{\prod_{j=1}^{k} x_{j|i}!} \left[\frac{\Gamma(a_1 + \dots + a_k)}{\Gamma(x_i + a_1 + \dots + a_k)} \prod_{j=1}^{k} \frac{\Gamma(x_{j|i} + a_j)}{\Gamma(a_j)}\right].
$$

The Evidence can be maximized over $a_1, \dots, a_k$ to get a good choice of the hyperparameters. Note that the factorial terms do not involve a_js so they can dropped in this maximization:

$$
\text{Evidence}(a_1, \dots, a_k) \propto \prod_{i=1}^{k} \left[\frac{\Gamma(a_1 + \dots + a_k)}{\Gamma(x_i + a_1 + \dots + a_k)} \prod_{j=1}^{k} \frac{\Gamma(x_{j|i} + a_j)}{\Gamma(a_j)}\right].
$$

It is always better to optimize the logarithm of the evidence (rather than the evidence directly):

$$
\begin{aligned}
&\log \text{Evidence}(a_1, \dots, a_k) \\
&= \text{constant} + \sum_{i=1}^{k} \left\{\log \Gamma(A) - \log \Gamma(x_i + A) + \sum_{j=1}^{k} \left[\log \Gamma(x_{j|i} + a_j) - \log \Gamma(a_j)\right]\right\}.
\end{aligned}
$$

where $A := a_1 + \dots + a_k$ and $x_i = \sum_{j=1}^{k} x_{j|i}$. One can compute the gradient with respect to a using the digamma function (see https://en.wikipedia.org/wiki/Digamma_function):

$$
\Psi(z) := \frac{d}{dz} \log \Gamma(z).
$$

This gives:

$$
\frac{\partial}{\partial a_j} \log \text{Evidence}(a_1, \dots, a_k) = \sum_{i=1}^{k} \left\{\Psi(A) - \Psi(x_i + A) + \Psi(x_{j|i} + a_j) - \Psi(a_j)\right\}.
$$

With these gradients (scipy has an inbuilt function for digamma calculation), some numerical optimization routine can be used for maximizing the log-Evidence over $a_1, \dots, a_k$ (see code).

References

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